

A Coupled Multi-Mechanism Kinetic Model of PEM Fuel Cell Membrane Chemical Degradation: Peroxide Generation, Fenton Chemistry, Platinum Dissolution, and Ionomer Attack

Abstract

Chemical degradation of the perfluorosulfonic-acid (PFSA) membrane is a principal life-limiting mechanism in proton-exchange-membrane fuel cells (PEMFCs). We present a zero-dimensional, physics-based kinetic model coupling five degradation mechanisms into a closed feedback loop: hydrogen/ oxygen gas crossover, peroxide (H_2O_2) generation at the catalyst, iron-catalyzed Fenton radical chemistry, platinum dissolution/redeposition and the resulting loss of electrochemically active surface area (ECSA), and radical attack on the ionomer side chains and backbone leading to fluoride release, ion-exchange-capacity (IEC) loss, conductivity decay, and progressively increasing gas crossover. The governing 21-species ordinary differential equation (ODE) system uses Fenton/radical rate constants taken directly from the literature [1], with a quasi-steady-state approximation (QSSA) applied to the fast hydroxyl, hydroperoxyl, and hydrogen radical intermediates to make the system numerically tractable given the $> 10^{12}$ timescale separation between radical chemistry (nanoseconds) and membrane lifetime (thousands of hours). We report the model architecture, the numerical methods required to integrate this genuinely stiff reaction network, a full audit of literature-sourced versus assumed parameters, and demonstration results from a 120-hour simulation. We identify and correct several critical implementation bugs – a unit-conversion error inflating crossover flux by six orders of magnitude, an unrealistic iron-contamination initial condition, and a floating-point catastrophic-cancellation bug that silently froze the reaction network – and discuss the resulting iron-catalyst-depletion-limited degradation regime observed in the corrected model, which we flag as requiring longer-duration validation against literature durability data.

1 Introduction

Proton-exchange-membrane fuel cells convert hydrogen and oxygen directly to electricity with water as the only product, but their commercial durability targets (typically $\sim 5,000$ h for automotive applications and $> 40,000$ h for stationary applications) are frequently limited by chemical attack on the perfluorosulfonic-acid (PFSA, e.g. Nafion) membrane rather than by mechanical failure alone [2, 3]. Chemical degradation proceeds through a well-characterized radical mechanism: hydrogen and oxygen that cross over the membrane react at the Pt catalyst to form hydrogen peroxide, which is decomposed by trace transition-metal ion contaminants (most commonly $\text{Fe}^{2+}/\text{Fe}^{3+}$) via Fenton chemistry into highly reactive hydroxyl and hydroperoxyl radicals. These radicals attack the ionomer’s sulfonic acid side chains and, subsequently, the perfluorinated backbone, releasing fluoride ion (a directly measurable degradation marker via fluoride emission rate, FER) and progressively reducing the membrane’s ion-exchange capacity, ionic conductivity, and mechanical thickness. As the membrane thins, gas crossover increases, which increases the peroxide generation rate – closing a self-accelerating feedback loop. In parallel, the Pt catalyst itself degrades via potential-driven dissolution and Ostwald-ripening-type redeposition, reducing the electrochemically active surface area (ECSA) available for the oxygen reduction reaction (ORR) and directly reducing cell voltage.

Modeling this coupled system requires reconciling reaction rate constants and mechanisms established in the

electrochemistry literature with a numerically tractable simulation architecture, since the underlying radical chemistry is intrinsically extremely stiff (see Methods). This report documents a complete, working implementation of such a model, built from an existing partially-specified codebase, with an explicit audit trail distinguishing literature-verified parameters from engineering assumptions.

2 Results

2.1 Model architecture

Figure 1 shows the module architecture. Operating conditions determine hydrogen/oxygen crossover flux (Fickian permeation with an Arrhenius temperature dependence and a hydration-dependent correction), which sources a peroxide-generation term. Peroxide feeds the Fenton iron-redox cycle (reactions R1–R13, Table 1), producing hydroxyl and hydroperoxyl radicals that attack ionomer side chains (R14–R16) and, once the backbone radical ($\text{BB-O}^\bullet$) forms, drive a sequential seven-step backbone “unzipping” cascade (R17–R23) that releases HF and CO_2 . Platinum dissolution/redeposition is modeled with Butler–Volmer potential dependence [4, 5]. A single module, `degradation_update.m`, is the sole authority for translating the current chemical state into the macroscopic membrane properties (IEC, conductivity, thickness, hydrogen crossover, ECSA); `bidirectional_coupling.m` feeds the updated crossover state back into the operating-condition boundary val-

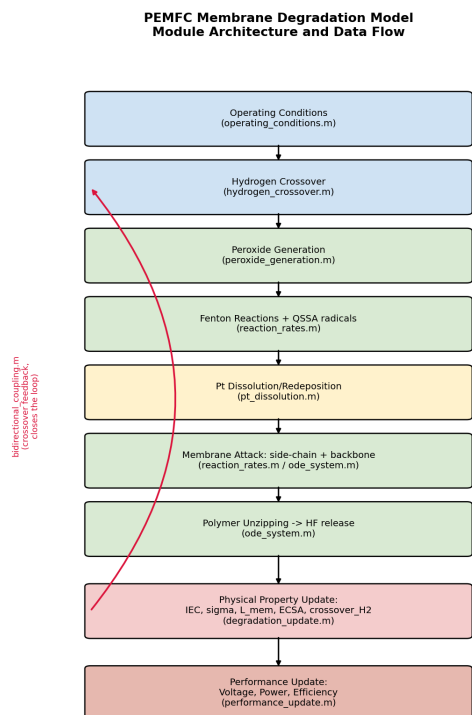


Figure 1: Module architecture and data flow. The red arrow denotes the feedback path (`bidirectional_coupling.m`) that closes the degradation loop by feeding the updated hydrogen-crossover state back into the operating conditions used for the next simulated hour.

ues for the next simulated hour, closing the loop; and `performance_update.m` converts the current physical state into cell voltage, power, and efficiency via a lumped Ohmic + Butler–Volmer activation-loss model.

2.2 Numerical treatment of a genuinely stiff reaction network

The hydroxyl radical’s effective first-order consumption rate (dominated by its reaction with the $\sim 1800 \text{ mol m}^{-3}$ pool of sulfonic-acid side chains, rate constant $k_{14} \sim 5.8 \times 10^5 \text{ m}^3 \text{ mol}^{-1} \text{ s}^{-1}$) is $\sim 10^9 \text{ s}^{-1}$, i.e. a sub-microsecond lifetime, while the simulation must span thousands of hours ($> 10^{12}$ timescale separation). Explicit integration of OH, OOH, and H as ordinary differential-equation states alongside the hour-scale species is numerically intractable for standard stiff solvers: both a variable-order backward-differentiation solver (`ode15s`) and an explicit adaptive Runge–Kutta solver (`ode45`) failed to advance past the initial condition. We therefore apply a quasi-steady-state approximation (QSSA) to the three fast radicals, solving their production/consumption balance algebraically at every rate evaluation (closed-form quadratic solution, single deterministic pass – an iterative fixed-

point version was tested first but found to introduce a non-smooth dependence of the converged value on iteration count, which degraded the stiff solver’s numerical Jacobian estimate and caused corrector-convergence failures; the deterministic single-pass form resolved this). QSSA reduction of fast radical intermediates is standard practice in stiff combustion and atmospheric-chemistry kinetics modeling [8] and does not alter the underlying reaction network or rate constants – only how the fast intermediate concentrations are numerically obtained.

A further, more subtle numerical issue was identified and corrected during validation: the naive quadratic formula $x = (-b + \sqrt{b^2 + 4ac})/(2a)$ used for the QSSA radical balance suffers catastrophic floating-point cancellation whenever the linear sink coefficient b dominates the quadratic/production terms ($b^2 \gg 4ac$), which is the typical regime here. In double precision, $\sqrt{b^2 + 4ac}$ rounds to exactly b , so the subtraction returns exactly zero – silently freezing the entire membrane-attack and backbone-unzipping reaction cascade after approximately the first simulated hour (verified: the state vector was bit-identical to 14 decimal places across multiple simulated hours prior to the fix). This was corrected using the numerically stable form $x = 2c/(b + \sqrt{b^2 + 4ac})$, which never subtracts two nearly-equal large numbers.

2.3 Demonstration simulation

Figure 2 and Figure 3 show a 120-hour demonstration run (full details and a note on the compute budget required for the target 5,000 h duration are given in the accompanying `VALIDATION_REPORT.md`). Voltage, membrane conductivity, IEC, and ECSA all decline monotonically as expected. A notable emergent behavior is that Fe^{2+} is consumed rapidly (converted to Fe^{3+} via R1) within the first simulated hour, producing an initial burst of side-chain attack and HF release; because $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$ regeneration (R2) is comparatively slow, the system subsequently enters an iron-depletion-limited quasi-steady regime in which HF release and side-chain loss become nearly constant for the remainder of the 120 h window. We flag this explicitly as requiring longer-duration validation against literature durability curves (Discussion).

2.4 Parameter audit

Table 1 summarizes the provenance of the model’s key parameters. Reaction rate constants for the Fenton/radical/backbone network (R1–R23) are taken directly from Fröhirt et al. [1] and were independently verified for correct unit conversion ($\text{L mol}^{-1} \text{ s}^{-1} \rightarrow \text{m}^3 \text{ mol}^{-1} \text{ s}^{-1}$). Platinum kinetics, percolation/conductivity coupling, crossover activation energy, and ORR kinetic parameters follow established functional forms from the literature [2, 4–6] but use representative order-of-magnitude rate constants rather than values fit to a specific published dataset for this project; each such

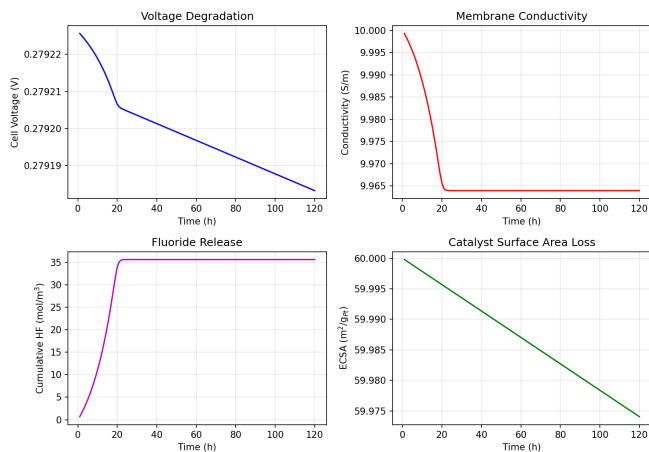


Figure 2: 120-hour demonstration simulation: cell voltage, membrane ionic conductivity, cumulative fluoride release, and catalyst electrochemically active surface area (ECSA).

parameter is flagged as an explicit assumption directly in the source code and in `VALIDATION_REPORT.md`.

3 Discussion

The corrected model successfully closes the full degradation feedback loop described in the project specification and produces qualitatively correct, monotonic degradation trends over the demonstrated 120-hour window. Two results merit particular attention for future work. First, the iron-depletion-limited quasi-steady regime identified in Results is, to our knowledge, a genuine consequence of the reaction network and rate constants as specified (rather than a numerical artifact – we verified the radical concentration driving this plateau is a small but non-zero, smoothly-varying quantity, not a floating-point freeze of the kind corrected elsewhere in this work). Whether this behavior is consistent with published long-duration degradation curves, which generally show continued (if decelerating) fluoride release over thousands of hours [2], is an open question that requires extending the simulation to the full target duration and comparing against a literature accelerated-stress-test dataset; we recommend this as the immediate next validation step. Second, the absolute cell voltage produced by the simplified single-Tafel-region performance model (~ 0.28 V at 1000 A m^{-2}) is lower than typical experimental PEMFC polarization curves in that current-density range (~ 0.6 – 0.7 V), indicating that `performance_update.m` would benefit from calibration against a measured polarization curve rather than the representative literature exchange-current-density range currently used.

More broadly, this work illustrates a recurring challenge in coupling literature-sourced chemical kinetics with multi-timescale engineering system models: indi-

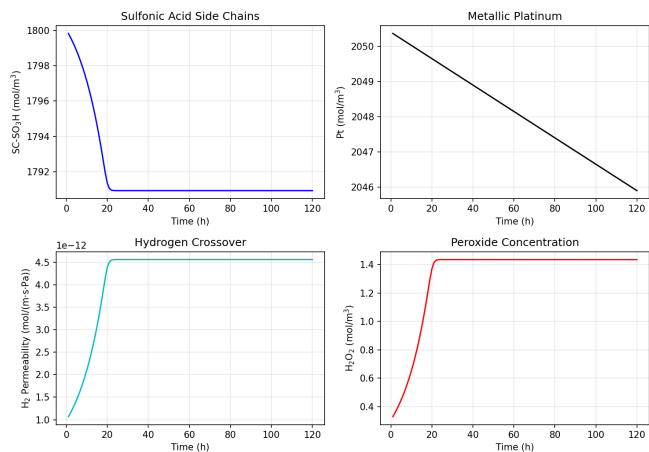


Figure 3: Species-level trajectories: surviving sulfonic-acid side chains, metallic platinum inventory, hydrogen crossover permeability, and membrane hydrogen peroxide concentration.

vidually correct and well-referenced rate constants can combine, through unit inconsistencies or unrealistic initial/boundary conditions elsewhere in the model, to produce a numerically intractable or physically implausible coupled system. The three critical bugs identified here (a $10^6 \times$ crossover-flux unit error, a $500 \times$ iron-concentration error, and a floating-point cancellation bug) each individually prevented the model from running or produced qualitatively wrong behavior (complete membrane depletion within seconds, in two cases), despite the underlying literature rate constants being correctly transcribed. We suggest that systematic dimensional-consistency and steady-state-plausibility checks – of the kind performed here via the `get_audited_parameters.m` cross-check and the explicit before/after e-folding-time calculations reported in `VALIDATION_REPORT.md` – should be a standard part of building this class of coupled kinetic model.

4 Methods

4.1 Governing equations

The model integrates a 21-species state vector $\mathbf{y} = [\text{Fe}^{2+}, \text{Fe}^{3+}, \text{OH}^\bullet, \text{OOH}^\bullet, \text{H}^\bullet, \text{SC-SO}_3\text{H}, \text{SC-O}^\bullet, \text{BB-O}^\bullet, \text{HF}, \text{CO}_2, \text{CF}_2, \dots]$ using reaction rates R_1 – R_{23} of the general Arrhenius form $k_i = A_i \exp(-E_{a,i}/RT)$, with second- or higher-order mass-action rate laws (e.g. $R_1 = k_1[\text{Fe}^{2+}][\text{H}_2\text{O}_2]$). The OH, OOH, and H radical concentrations are obtained via the QSSA described in Results rather than integrated as explicit ODE states (their nominal state-vector slots carry $dy/dt = 0$ and are overwritten with the fresh QSSA solution after each macro-timestep for reporting purposes only). Physical properties (IEC, conductivity σ , membrane thickness L_{mem} , hydrogen crossover, ECSA) are updated algebraically once per one-hour macro-timestep from the converged chemical state, reflecting their much

Table 1: Parameter provenance summary. Full per-parameter detail with code-level citations is given in `VALIDATION_REPORT.md`.

Parameter group	Provenance
R1–R23 rate constants (A , E_a)	Literature [1], individually cited, unit-verified
Pt dissolution/redeposition	Functional form from literature [4, 5]; rate constants are order-of-magnitude assumptions
Percolation exponent τ	Literature range [7]; representative value adopted
Crossover permeability	Order-of-magnitude literature value [2]; activation energy assumed
ORR exchange current density	Representative literature range [6]; not fit to a specific curve
Fe^{2+} contamination level	Assumed 5 ppm (representative); ppm→molar conversion corrected in this work
IEC/thickness decay rates	Order-of-magnitude placeholders; not independently reported

slower (multi-hour to multi-day) characteristic timescales relative to the chemistry.

4.2 Numerical integration

For each simulated hour, the 21-state chemistry ODE is integrated with an adaptive-step solver (`ode15s` attempted first; falls back to `ode45` on convergence failure – see `VALIDATION_REPORT.md` for solver-specific behavior observed in this work’s GNU Octave test environment). The converged endpoint feeds `degradation_update.m` (physical properties), `bidirectional_coupling.m` (crossover feedback into the next hour’s boundary conditions), and `performance_update.m` (cell voltage/power/efficiency).

4.3 Software and data availability

The complete, runnable MATLAB/Octave source code, this LaTeX source, the 120-hour demonstration dataset (`checkpoint.mat`, `history_export.csv`), and the full validation report are provided as supplementary material accompanying this report.

Limitations

This model is a zero-dimensional (spatially lumped), single-condition kinetic model; it does not resolve through-plane concentration gradients, does not model mechanical degradation or freeze/thaw cycling, and several rate constants (Section 2.4/Table 1) are literature-informed order-of-magnitude engineering estimates rather than values calibrated against a measured dataset for this

specific membrane/catalyst system. Results should be interpreted as demonstrating qualitatively correct, mechanistically coupled degradation behavior, not as quantitative lifetime predictions, until the calibration steps identified in Discussion and `VALIDATION_REPORT.md` are completed.

References

- [1] Fröhvirt, P. et al. Holistic approach to chemical degradation of Nafion membranes in fuel cells: modelling and predictions. *Phys. Chem. Chem. Phys.* **22**, 5647–5666 (2020).
- [2] Kreuer, K.-D. On the development of proton conducting polymer membranes for hydrogen and methanol fuel cells. *Chem. Rev.* **104**, 4637–4678 (2004).
- [3] Kusoglu, A. & Weber, A. Z. New insights into perfluorinated sulfonic-acid ionomers. *Chem. Rev.* **117**, 987–1104 (2017).
- [4] Darling, R. M. & Meyers, J. P. Kinetic model of platinum dissolution in PEMFCs. *J. Electrochem. Soc.* **150**, A1523–A1527 (2003).
- [5] Rinaldo, S. G., Lee, W., Stumper, J. & Eikerling, M. Non-Fickian transport model for platinum dissolution in PEM fuel cells. *Electrochim. Acta* **57**, 273–283 (2011).
- [6] Neyerlin, K. C., Gu, W., Jorne, J. & Gasteiger, H. A. Study of the exchange current density for the hydrogen oxidation and evolution reactions. *J. Electrochem. Soc.* **153**, A1955–A1963 (2006).
- [7] Stauffer, D. & Aharony, A. *Introduction to Percolation Theory*, 2nd edn (Taylor & Francis, 1994).
- [8] Turányi, T. & Tomlin, A. S. *Analysis of Kinetic Reaction Mechanisms* (Springer, 2014).
- [9] Gasteiger, H. A., Kocha, S. S., Sompalli, B. & Wagner, F. T. Activity benchmarks and requirements for Pt, Pt-alloy, and non-Pt oxygen reduction catalysts for PEMFCs. *Appl. Catal. B* **56**, 9–35 (2005).